

(2-d)-kernels in Graphs and Digraphs

From one counting argument to a 755,516-graph census

Siddharth Narayan Mishra

August 25, 2026

Setup

Richardson's Theorem

A **kernel** of a digraph D is a set $S \subseteq V$ that is

- **independent/stable:** no arc, in either direction, joins two vertices of S ;
- **dominating/absorbent:** every $v \notin S$ has an arc to *some* vertex of S .

Theorem 1 (Richardson, 1953)

Every digraph with no odd directed cycle has a kernel.

The natural harder question

Richardson's domination condition is *one* arc into S .

What if every vertex outside S needs **two** distinct neighbours in S , not just one?

That one change is the entire subject of this talk.

Definition: the (2-d)-kernel

(2-d)-kernel

A set $S \subseteq V$ is a **(2-d)-kernel** of a digraph $D = (V, A)$ if

- (i) S is independent, and
- (ii) every $v \notin S$ has $|N^+(v) \cap S| \geq 2$ (**2-absorbent**).

An undirected graph is treated as a **symmetric digraph**.

A. Wloch, *On 2-dominating kernels in graphs*, Australas. J. Combin. **53** (2012) 273–284.

- **Theorem 2.4** a bipartite graph with all pendant vertices at pairwise even distance has a $(2-d)$ -kernel built from one bipartition class.
- **Theorem 2.11** a characterization of when *simplicial graphs* have a $(2-d)$ -kernel.

The Follow-Ups

- Bednarz, Hernández-Cruz, Wloch, *On the existence and the number of (2-d)-kernels in graphs*, Ars Combin. **121** (2015) 341-351 : attempts NP-completeness for general graphs.
- Bednarz, Wloch, *An algorithm determining (2-d)-kernels in trees*, Util. Math. (2017) : a linear algorithm, restricted to trees.

Every published result above is about **undirected graphs**.

Before this project, no digraph version of 2-domination existed in the literature at all.

First Principles

Theorem 2: Bipartite Digraphs

Theorem 2

If the underlying graph of a digraph D is bipartite with parts P, Q , and every vertex of Q has $d^+ \geq 2$, then P is a $(2-d)$ -kernel.

Proof.

Bipartiteness means every arc runs between P and Q , so P is automatically independent (no arc stays inside P). Every $v \in Q$ has all its out-arcs landing somewhere, and none of them can land back inside Q , so all of v 's out-arcs land in P — at least 2 of them. □

Theorem 3

Theorem 3

Every strongly connected digraph with no odd directed cycles and $\delta^+(D) \geq 2$ has two disjoint $(2, d)$ -kernels.

Cycle Decomposition

The proof needs one lemma first: every closed directed walk decomposes into directed cycles.

Lemma 4

Let $W = w_0, w_1, \dots, w_L$ ($w_0 = w_L$) be a closed directed walk of length L . Then there exist directed cycles $C_1, \dots, C_m$ of D with $|C_1| + \dots + |C_m| = L$.

Cycle Decomposition Proof

Proof.

Strong induction on L .

Base case: $L = 0$.

$$m = 0.$$

Inductive step: Let $L \geq 1$.

There are two cases:



Cycle Decomposition Proof

Proof.

Case 1: $w_0, \dots, w_{L-1}$ are all distinct.

$W = w_0, w_1, \dots, w_{L-1}, w_0$ is a simple cycle of length L .



Cycle Decomposition Proof

Proof.

Case 2: Some vertex is repeated.

Choose

$$0 \leq a < b \leq L - 1, \quad w_a = w_b.$$

Then W splits into two shorter closed walks:

$$W_1 = w_a, \dots, w_b, \quad |W_1| = b - a,$$

$$W_2 = w_0, \dots, w_a, w_{b+1}, \dots, w_L, \quad |W_2| = L - (b - a).$$



Cycle Decomposition Proof

Proof.

Since

$$|W_1|, |W_2| < L,$$

the induction hypothesis gives cycle decompositions of both.

Combining them gives a decomposition of W , with

$$|W_1| + |W_2| = (b - a) + L - (b - a) = L.$$



The Parity Claim

Claim. Every directed path from r to a vertex v has the same length parity.

Thus we may define

$$\text{par}(v) := |P| \pmod{2}$$

for any directed path $P : r \rightarrow v$.

It remains to show that this is well-defined.

The Parity Claim

Proof.

Suppose $P, Q : r \rightarrow v$ have different parity:

$$|P| \not\equiv |Q| \pmod{2}.$$

By strong connectivity, choose a directed path

$$R : v \rightarrow r.$$

Then $P + R$ and $Q + R$ are closed walks, with

$$|P + R| = |P| + |R|, \quad |Q + R| = |Q| + |R|.$$



The Parity Claim

Proof.

By Lemma 4, both closed walks are sums of even cycles.
Hence both lengths are even:

$$|P| + |R| \equiv |Q| + |R| \equiv 0 \pmod{2}.$$

Therefore

$$|P| \equiv |Q| \pmod{2},$$

a contradiction.

$\text{par}(v)$ is well-defined.



Building the Two Kernels

Let V_0, V_1 be the two parity classes.

Every arc $u \rightarrow v$ flips parity (parity of r - u path and r - v path will be different)

So V_0, V_1 are independent, and every out-neighbour of a V_i -vertex lies in V_{1-i} .

$$\delta^+(D) \geq 2$$

Both V_0 and V_1 are disjoint (2-d)-kernels. ■

Send in the Machines

Built Systematic Tooling

Instead of trusting further hand-picked examples, built a tool that can

- generate *every* graph or digraph up to a given size, exhaustively,
- verify (2-d)-kernels against the definition directly, with no shortcuts,
- and try a cheap propagation method first, so that when it fails, the failure itself is data.

Tool Architecture

Every function that returns a candidate set passes it through a single verifier, `is_2kernel`, before returning it: check independence, then check every outside vertex has ≥ 2 out-arcs into the set.

A (2-d)-kernel is always a **maximal independent set** of the underlying graph (anything outside it is already adjacent to it). So a **complete, obviously-correct** reference solver exists: enumerate every maximal independent set (Bron-Kerbosch), keep the ones `is_2kernel` accepts.

Exponential in the worst case, but it is the ground truth every faster method is checked against, and it is cheap enough for the exhaustive ranges used here.

The Forcing Closure

Before full search, a cheap propagation is tried first. Each vertex is **IN**, **OUT**, or **UNKNOWN**; rules apply repeatedly until nothing changes.

$$U(v) := N^+(v) \setminus \text{OUT}$$

- R0 $N^+(v)$ has no independent pair $\Rightarrow v$ is **IN**.
- R1 v **IN** $\Rightarrow$ every underlying neighbour of v is **OUT**.
- R2 v **OUT**, and $U(v)$ has no independent pair $\Rightarrow$ **CONFLICT**.
- R3 v **OUT** and $|U(v)| = 2 \Rightarrow$ every member of $U(v)$ is **IN** and $|U(v)| < 2$ is a **CONFLICT**.
- R4 v **UNKNOWN**, and R2's test on v 's current non-**OUT** out-neighbours would conflict $\Rightarrow v$ is **IN**.

Why any of this is a proof, not a guess

Each rule only ever labels v from information already established. The question this must answer: if the closure says **CONFLICT**, how do we *know* no (2-d)-kernel exists, rather than just having failed to find one?

Soundness

For every (2-d)-kernel S that might exist, every label the closure ever assigns agrees with S — so a labelling that becomes self-contradictory proves S cannot exist.

Lemma: Soundness of the Rules

Lemma 5 (Soundness)

Suppose D has a (2-d)-kernel S . At every stage of the closure,

$$v \in \mathbf{IN} \Rightarrow v \in S, \quad v \in \mathbf{OUT} \Rightarrow v \notin S.$$

*Consequently, if the closure ever reaches **CONFLICT**, D has **no** (2-d)-kernel at all.*

Soundness Proof: R0 and R1

Proof.

Induction on the number of rule firings; assume the claim for every label set so far.

R0: if $N^+(v)$ has no independent pair and $v \notin S$, domination needs two independent members of $N^+(v)$ in S . This is impossible. So $v \in S$.

R1: if $v \in \mathbf{IN}$, the hypothesis gives $v \in S$; any underlying neighbour u of v cannot also lie in S (independence), so $u \notin S$, matching **OUT**. □

Soundness Proof: R2 and R3

Proof.

Both rules test $U(v) := N^+(v) \setminus \text{OUT}$. By hypothesis every vertex already **OUT** is not in S , so $N^+(v) \cap S \subseteq U(v)$. If v is **OUT** (so $v \notin S$, by hypothesis), domination needs $|N^+(v) \cap S| \geq 2$ — two *independent* members of $U(v)$ inside S .

R2: $U(v)$ has no independent pair at all, so no such S exists: **CONFLICT** is correct.

R3: if $|U(v)| = 2$, both must lie in S (only 2 candidates, need 2); if $|U(v)| < 2$, impossible, **CONFLICT**. □

Soundness Proof: R4, and closing the induction

Proof.

R4: v is **UNKNOWN**. Suppose $v \notin S$: exactly as in R2, domination would need an independent pair inside $U(v)$; the rule only fires when that test already conflicts, i.e. no such pair exists. So $v \notin S$ is impossible, forcing $v \in S$ to match the new **IN** label.

Every rule preserves the invariant whenever it fires, so it holds at every stage, including the fixed point. □

The Three Verdicts, Now Justified

Running the rules to a fixed point gives one of three verdicts, each now backed by Lemma 5:

- **CONFLICT** Lemma 5 shows “ D has a (2-d)-kernel” is self-contradictory: **no (2-d)-kernel exists**;
- **SOLVED** every vertex decided, and the **IN** set is verified as an actual (2-d)-kernel; Lemma 5 shows any other kernel would have to agree with every forced label, so it is the **only** one;
- **UNDECIDED** some vertices left unresolved; the closure simply didn't finish the job.

One `sqlite` file, keyed by a **canonical** graph6/digraph6 string from `pynauty`; so isomorphic duplicates can never split into two rows, and every count reported later is a count of isomorphism classes, not labelled instances.

Exhaustive generation past small n uses **canonical augmentation**: extend every graph on $n-1$ vertices by one new vertex in every possible way, dedup by certificate.

Chordal Graphs

Chordal graphs need **two** rules to fully decide:

- **Spread:** (R1) if v is **IN**, mark every neighbour **OUT**.
- **Force:** (R0 and R4) if v is **UNKNOWN** and its non-**OUT** neighbours contain no independent pair, mark v **IN**.

The Shortcut Never Fails

Across every chordal graph on at most 9 vertices (**17,175 graphs, exhaustive**) Spread and Force together decide the graph completely.

Zero **UNDECIDED** verdicts. Maximum (2-d)-kernel count: 1.

Why?

Simplicial Vertices?

Definitions

A vertex x is **simplicial** if $N(x)$ is a clique. A graph is **chordal** if it has no induced cycle of length ≥ 4 .

Two facts:

- every non-empty chordal graph has a simplicial vertex;
- every induced subgraph of a chordal graph is itself chordal.

Theorem 6

Theorem 6

*On a chordal graph, Spread and Force run to a fixed point never leave a vertex **UNKNOWN**. Consequently a chordal graph has at most one (2-d)-kernel, and (2-d)-kernel is solvable in polynomial time on chordal graphs.*

The general problem is NP-complete.

Theorem 6 Proof

Proof.

Suppose some vertex is **UNKNOWN** at the fixed point. Let

$$W = \{v \in V(G) : v \text{ is UNKNOWN}\} \neq \emptyset.$$

For every $v \in W$,

$$N(v) \cap \mathbf{IN} = \emptyset,$$

since otherwise **Spread** would have fired.

Hence the current non-**OUT** neighbourhood of v is

$$N(v) \cap W.$$



Theorem 6 Proof

Proof.

Since G is chordal, its induced subgraph $G[W]$ is also chordal.

As $W \neq \emptyset$, $G[W]$ has a simplicial vertex x . Thus

$$N_{G[W]}(x) = N(x) \cap W$$

is a clique.

But $N(x) \cap W$ is exactly the current non-**OUT** neighbourhood of x . Therefore **Force** fires on x .

This contradicts the assumption that we are at a fixed point.



Uniqueness

Suppose S is *any* (2-d)-kernel of G

Lemma 5 gives, simultaneously,

$$\mathbf{IN} \subseteq S \quad \text{and} \quad S \subseteq V \setminus \mathbf{OUT}.$$

Since $W = \emptyset$, every vertex is **IN** or **OUT**, so $V \setminus \mathbf{OUT} = \mathbf{IN}$. The two inclusions sandwich S :

$$\boxed{\mathbf{IN} \subseteq S \subseteq \mathbf{IN} \implies S = \mathbf{IN}}$$

This holds for *whatever* S was. So every (2-d)-kernel of G , if one exists, must literally equal **IN**.

Just check if the **IN** set is actually a (2-d)-kernel. If it is, it is the only one; if it is not, no (2-d)-kernel exists.

For the complexity claim: chordal recognition and a simplicial elimination ordering are polynomial; each rule firing is a polynomial-time clique/neighbour check; the process fixes one previously-**UNKNOWN** vertex per firing, so it terminates in at most n firings; the final check of **IN** is a single linear pass.

For every chordal graph G , there exists a polynomial-time algorithm that decides whether G has a $(2,d)$ -kernel. Moreover, if such a kernel exists, it is unique.

Generalize Down

Try Degeneracy

Definition

A graph is ***k*-degenerate** if it can be reduced to nothing by repeatedly removing a vertex of degree $\leq k$. The **degeneracy** is the smallest such k (Matula-Beck peeling).

Every chordal graph admits a perfect elimination ordering, which gives a vertex of minimum degree at each step.

So the natural next question is: does the chordal argument survive when we replace **chordality** by the weaker condition of **bounded degeneracy**?

Degeneracy 0 $\rightarrow$ edgeless

Degeneracy 1 $\rightarrow$ forests

Works at Degeneracy 1

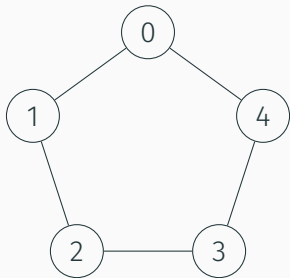
degeneracy	graphs	forcing disagreements	max #(2-d)-kernels
0 (edgeless)	10	0	1
1 (forests)	299	0	1

Forcing is complete at degeneracy ≤ 1 . But 1-degenerate graphs are exactly the forests (2017 Paper)

Breaks at Degeneracy 2

degeneracy	graphs	forcing disagreements	max #(2-d)-kernels
2	35,739	3,388 (9.5%)	4

The Counterexample: C_5



Why Low Degree Is Not Simplicial

The chordal proof uses the following fact: at a fixed point, the undecided region is chordal, and hence contains a *simplicial* vertex.

Degeneracy 2 guarantees a vertex of degree at most 2 at every stage of peeling, but does not guarantee that its neighbourhood is a clique.

Thus, low degree does not imply simpliciality.

Generalize Sideways

Strengthen the object instead

Weakening the graph class (chordal $\rightarrow$ degenerate) just broke. Try the other direction: keep the graph class, **strengthen the object** from a graph to a digraph.

Question: if the *underlying graph* of a digraph D is chordal, does the Theorem 6 argument still decide D ?

Key Point

Simpliciality concerns the full neighbourhood $N(v)$, not merely the forward neighbourhood $N^+(v)$. $N^+(v) \subseteq N(v)$ so Force still works.

Spread and Force, on D , stated with this notation

At every stage every $v \in V$ has a status in $\{\mathbf{IN}, \mathbf{OUT}, \mathbf{UNKNOWN}\}$; write $\mathbf{IN}, \mathbf{OUT}, \mathbf{UNKNOWN}$ also for the three sets of vertices.

Spread $v \in \mathbf{IN} \Rightarrow N(v) \subseteq \mathbf{OUT}$. Uses N (the underlying graph): an arc *either way* between two vertices of S breaks independence, so both directions must be swept.

Force $v \in \mathbf{UNKNOWN}$ and $N^+(v) \setminus \mathbf{OUT}$ has no independent pair $\Rightarrow v$ moves to $\mathbf{IN}$. Uses N^+ (the digraph): 2-absorbency, $|N^+(v) \cap S| \geq 2$, only ever counts out-arcs.

These are exactly rules R1 and R0/R4 from the general closure, specialised to their relevant graph.

Theorem 7

Theorem 7

*If the underlying graph G of a digraph D is chordal, then Spread and Force run to a fixed point never leave a vertex **UNKNOWN** on D . Consequently D has at most one $(2-d)$ -kernel, and $(2-d)$ -kernel is solvable in polynomial time whenever G is chordal with **no restriction on $\delta^+(D)$ at all.***

Theorem 7 Proof: what a fixed point knows

Proof.

Suppose that some vertex is **UNKNOWN**. Let

$$W := \{v \in V : v \in \mathbf{UNKNOWN}\} \neq \emptyset.$$

Fix $v \in W$. If some $u \in N(v)$ had $u \in \mathbf{IN}$, Spread applied to u would place $v \in \mathbf{OUT}$. This contradicts $v \in \mathbf{UNKNOWN}$. Hence

$$N(v) \cap \mathbf{IN} = \emptyset \quad \text{for every } v \in W.$$



Theorem 7 Proof: passing from N to N^+

Proof.

V splits into **IN**, **OUT**, W . For $v \in W$, the previous frame gives $N(v) \cap \mathbf{IN} = \emptyset$, so

$$N(v) \setminus \mathbf{OUT} = N(v) \cap W.$$

Now use $N^+(v) \subseteq N(v)$: intersecting both sides above with $N^+(v)$ gives $N^+(v) \cap \mathbf{IN} = \emptyset$ as well, so the *same* splitting argument applied to $N^+(v)$ gives

$$N^+(v) \setminus \mathbf{OUT} = N^+(v) \cap W \quad \text{for every } v \in W.$$



Theorem 7 Proof: a simplicial vertex in W

Proof.

All of the above is about D ; chordality is a hypothesis about G alone. G is chordal, so its induced subgraph $G[W]$ is chordal too (induced subgraphs of chordal graphs are chordal). As $W \neq \emptyset$, $G[W]$ has a simplicial vertex $x \in W$: its neighbourhood inside $G[W]$,

$$N_{G[W]}(x) = N(x) \cap W,$$

is a clique in G , i.e., $N(x) \cap W$ contains no independent pair. □

Theorem 7 Proof: Force fires on x

Proof.

Apply the identity from two frames ago at $v = x \in W$:

$$N^+(x) \setminus \mathbf{OUT} = N^+(x) \cap W.$$

Since $N^+(x) \subseteq N(x)$, this set is a *subset* of $N(x) \cap W$, which the previous frame showed is a clique in G . A subset of a clique contains no independent pair, so

$N^+(x) \setminus \mathbf{OUT}$ has no independent pair.

As $x \in \mathbf{UNKNOWN}$, this is exactly Force's hypothesis at x : Force would move x to $\mathbf{IN}$. This is contradiction. Hence $W = \emptyset$. □

Uniqueness: the same sandwich, on D

Let $S^* := \mathbf{IN}$. Suppose S is *any* (2-d)-kernel of D .

Lemma 5 gives $\mathbf{IN} \subseteq S \subseteq V \setminus \mathbf{OUT}$, and $W = \emptyset$ makes $V \setminus \mathbf{OUT} = \mathbf{IN}$:

$$\boxed{\mathbf{IN} \subseteq S \subseteq \mathbf{IN} \implies S = \mathbf{IN} = S^*}.$$

Existence: one check settles it

One direct check of S^* against the definition.

Theorem 7

*If the underlying graph of D is chordal, the forcing closure decides D completely: at most one $(2-d)$ -kernel, polynomial time with **no restriction on δ^+** at all.*

Oriented Graphs

Definition

An **oriented graph** has no **digons**: for any two vertices, at most one of $u \rightarrow v$, $v \rightarrow u$ is present.

Question: with $\delta^+ \geq 2$, how small can a (2-d)-kernel-having oriented graph be?

Let D be oriented, $\delta^+(D) \geq 2$, and suppose D has a (2-d)-kernel J .

- $j = |J|$, $B = V \setminus J$, $t = |B| = n - j > 0$.
- For $x \in J$, let d_x = the number of B -vertices with an arc *into* x .

First inequality: 2-domination

Every vertex of B sends ≥ 2 arcs into J (that is what 2-domination means for J). Summing over J , counting each such arc once by its J -endpoint:

$$\sum_{x \in J} d_x \geq 2t.$$

Second inequality: independence + no digons

Fix $x \in J$. Since J is independent, every out-arc of x goes to B .

Since D has **no digons**, x cannot send an out-arc to any vertex of B that *already* points at x , so x 's out-arcs are confined to the $t - d_x$ vertices of B that do *not* point at x .

Since $d^+(x) \geq \delta^+(D) \geq 2$:

$$t - d_x \geq 2 \quad \implies \quad d_x \leq t - 2, \quad \text{for every } x \in J.$$

$$2t \leq \sum_{x \in J} d_x \leq j(t-2).$$

This forces $t \geq 3$ (otherwise $t-2 \leq 0$ and no positive j can satisfy it), and

$$j \geq \frac{2t}{t-2}.$$

Minimize $n = j + t$

t	smallest $j \geq 2t/(t-2)$	$n = j + t$	
3	6	9	
4	4	8	minimum
5	4	9	
6	3	9	
7	3	10	rising from here

For $t \geq 6$, $n \geq t + 3$, which only grows $\rightarrow$ no larger t beats $n = 8$.

Theorem 8

No oriented graph on fewer than 8 vertices with $\delta^+ \geq 2$ has a (2-d)-kernel. $n = 8$ is attained, and only at $t = j = 4$.

So $\delta^+ \geq 2$, which *helps* build a (2-d)-kernel with digons available (Theorems 2 and 3), actively **fights against** one once digons are forbidden.

DAGs

Theorem 9: DAGs

Theorem 9

A DAG has at most one $(2, d)$ -kernel, and its existence can be decided in linear time.

Fix a topological ordering

$$v_1, \dots, v_n$$

and process the vertices in reverse order.

When v_i is processed, every out-neighbour of v_i has already received a permanent status. Let

$$k = |\{u \in N^+(v_i) : u \text{ is IN}\}|.$$

Theorem 9: Proof

The three cases are:

$$k = 0 \Rightarrow v_i := \mathbf{IN},$$

$$k = 1 \Rightarrow \text{no } (2, d)\text{-kernel exists,}$$

$$k \geq 2 \Rightarrow v_i := \mathbf{OUT}.$$

Theorem 9: Proof

Proof.

If $k = 0$, assigning $v_i \in S$ is consistent with independence, since no out-neighbour of v_i is in S .

If $k = 1$, let u be the unique **IN** out-neighbour of v_i . Then

$$v_i \rightarrow u, \quad u \in S.$$

Hence $v_i \in S$ violates independence, while $v_i \notin S$ violates the requirement that every **OUT** vertex have at least two out-neighbours in S . Thus no $(2, d)$ -kernel exists.

If $k \geq 2$, $v_i \notin S$ is forced, since putting v_i in S would violate independence with an **IN** out-neighbour. Moreover, v_i is already 2-dominated by its $k \geq 2$ out-neighbours in S . □

Theorem 9: Completion

If the sweep terminates without encountering $k = 1$, define

$$S = \{v_i : v_i \text{ was assigned } \mathbf{IN}\}.$$

Every **IN** vertex was assigned when $k = 0$, so no directed edge joins two vertices of S . Hence S is independent.

Every **OUT** vertex was assigned when $k \geq 2$, so it has at least two out-neighbours in S . Hence S 2-dominates $V(D) \setminus S$.

Theorem 9: Completion

Therefore S is a $(2, d)$ -kernel.

Every status is forced by the already processed out-neighbours. Hence no branching is possible, and if a $(2, d)$ -kernel exists, it is unique.

The reverse topological sweep examines each vertex and its out-neighbours once, giving an $O(|V| + |E|)$ algorithm.